DEEP LEARNING — FINAL PROJECT REPORT

Sleep Apnea Detection via Respiratory & Oximetry Signals

A Comparative Study: CNN-Based Deep Learning vs. Classical Machine Learning

- Samuel Braga Marques
- Miguel De Menezes Coelho
- Popescu Ianis-Stephan
- Juan Francisco Cortez Hervas

June 2026

PRESENTATION OVERVIEW

Four Stops Through the Project

Introduction

Why automatic apnea detection matters — and the dataset behind this study

Methods

Two parallel pipelines: an end-to-end 1D CNN and a feature-based ExtraTreesClassifier

Results

Test performance, training curves, and Leave-One-Subject-Out cross-validation

Discussion

What the comparison reveals, where it falls short, and what comes next

INTRODUCTION

Why Automatic Apnea Detection?

- Obstructive Sleep Apnea (OSA): repeated cessation of breathing during sleep → oxygen desaturation, fragmented sleep, long-term cardiovascular & metabolic risk
- Gold standard — Polysomnography (PSG) — is expensive, inconvenient, and resource-intensive, leaving OSA substantially under-diagnosed
- Research question: can ML models (deep & classical) reliably tell normal breathing apart from apnea using cheaper, non-invasive sensors?

Class 0 — **Normal**

Regular, unobstructed breathing

Class 1 — **Apnea ≈ 10s**

Short-duration apnea event

Class 2 — **Apnea ≈ 20s**

Longer-duration apnea event

INTRODUCTION

Dataset & Sensor Modalities

Respiratory Oximetry Apnoea dataset — PhysioNet · induced apnea events under controlled lab conditions

20

Healthy adult subjects

3

PEEP levels: 0 / 4 / 8 cmH_2O

~250 Hz

Sampling rate, both sensors

Inline Mask Sensor

Inline_PQ_Data

Spirometry-grade pressure mask. 2 channels: Gauge Pressure & Inspiratory Differential Pressure. Apnea → sustained low-amplitude, near-flat segments.

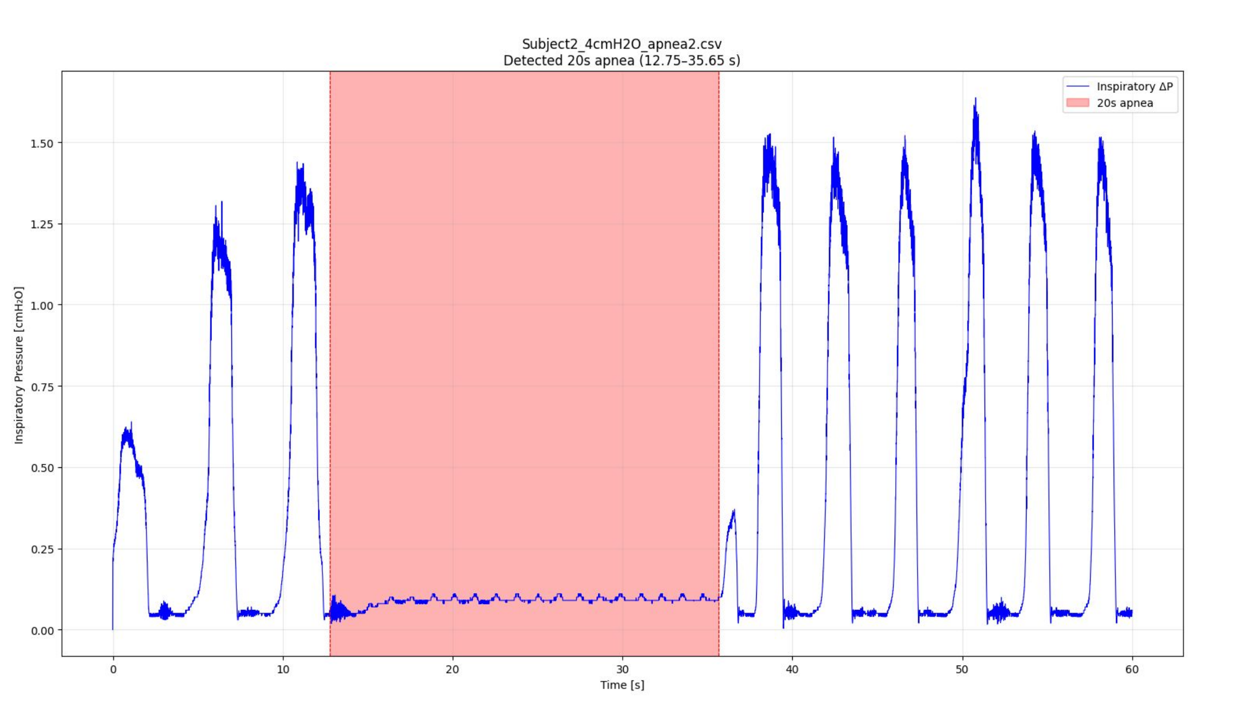
Neck Pulse Oximeter

Neck_Pulse_Oximeter_Data

Custom neck-worn PPG sensor. 8 channels: 4 Red (PD1–PD4) + 4 Infrared (PD1_9–PD4_9). Captures pulse waveform & respiratory-modulated pulsatility.

Data from each Sensor Modality

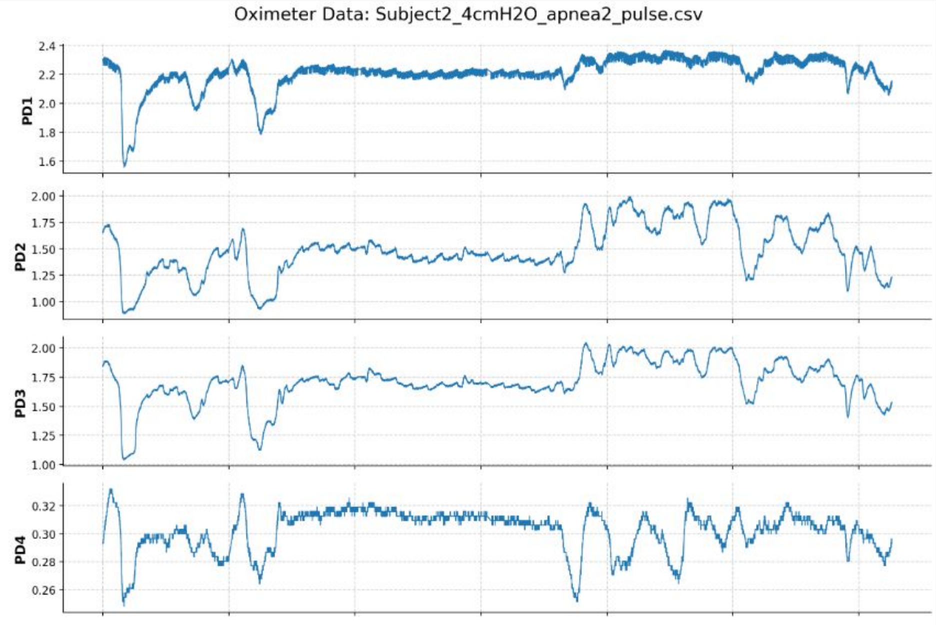
Inline Mask

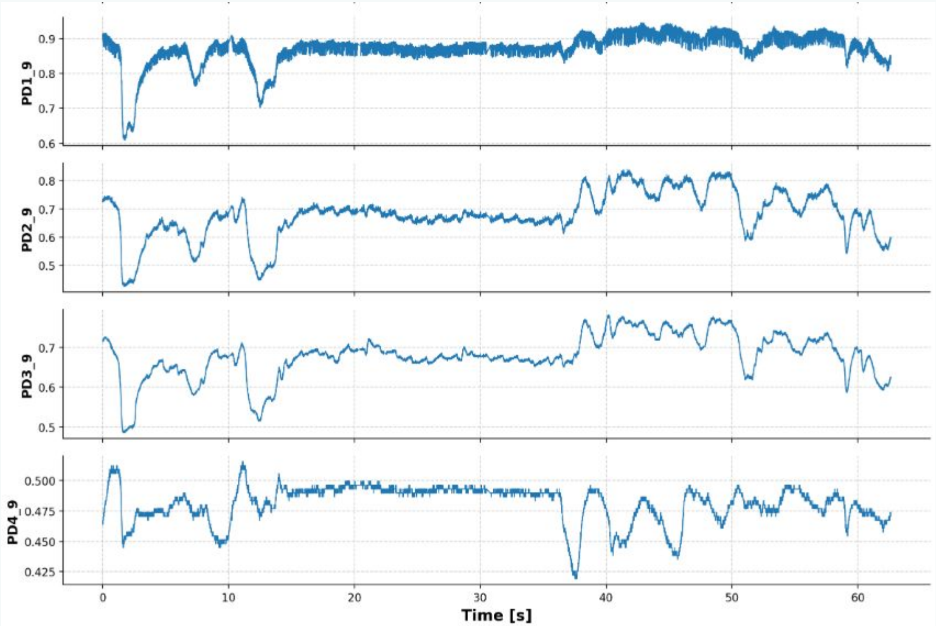


INTRODUCTION

Data from each Sensor Modality

Neck Sensor Oximeter





METHODS

Preprocessing & Windowing Pipeline

1

Resampling

Uniform 250 Hz grid across all subjects & tasks

→

2

Filtering

4th-order Butterworth (filtfilt) — 2.5 Hz mask / 5 Hz oximeter cutoff

→

3

Apnea Labeling

Rolling-std detector matches event duration to filename metadata

→

4

Windowing

Sliding windows; subject-level split (1–14 / 15–17 / 18–20)

Class Imbalance: normal-class windows far outnumber apnea windows. CNN ⟶ weighted cross-entropy from sklearn's compute_class_weight. ExtraTreesClassifier⟶ class_weight='balanced'.

METHODS — MODEL 1

1D CNN: ApneaCNN1D Architecture

Conv Block 1

32 filters, k=15 ⟶ BN ⟶ ReLU ⟶ Dropout1d(0.10) ⟶ MaxPool(2)

Conv Block 2

64 filters, k=9 ⟶ BN ⟶ ReLU ⟶ Dropout1d(0.15) ⟶ MaxPool(2)

Conv Block 3

128 filters, k=5 ⟶ BN ⟶ ReLU ⟶ Dropout1d(0.20) ⟶ MaxPool(2)

Conv Block 4 (Bottleneck)

64 filters, k=3 ⟶ BN ⟶ ReLU ⟶ Dropout1d(0.20) ⟶ MaxPool(2)

Global Pool + FC Head

AdaptiveAvgPool1d(16) ⟶ Flatten ⟶ 128 ⟶ ReLU ⟶ Dropout(0.4) ⟶ 3 classes

Design

- Kernels shrink 15⟶9⟶5⟶3 to keep consistent real-world receptive field as MaxPool halves resolution
- Dropout1d zeroes entire feature channels — best granularity for 1D conv features
- Adam lr=1e-3, weight decay 5e-4, ReduceLROnPlateau scheduler
- Label smoothing 0.05 + WeightedRandomSampler fight overconfidence & class imbalance
- Binary: full 60 s window. Multiclass: 45 s windows, stride 2 s overlap during training

Classical Baseline: ExtraTreesClassifier Pipeline

Feature Engineering

w00_10	w10_25	w25_40
w40_60	w00_30	full

12 statistical features per channel per window: mean, std, median, p05, p95, iqr, range, rms, mad, diff_abs_mean, diff_std, slope. Wider windows add FFT spectral features (dominant freq, resp band, cardiac band). ExtraTrees was selected because it handles nonlinear tabular features, does not require scaling, trains quickly on CPU, and provides feature importances.

sklearn Pipeline

1. SimpleImputer (strategy='median'): handles NaN from short windows
2. ExtraTreesClassifier: 600 trees, max_features='sqrt', class_weight='balanced'
3. PEEP level excluded: keeps model signal-based, prevents protocol leakage

Evaluation: Leave-One-Subject-Out CV (LOSO)

20 folds — one subject held out per fold. Simulates real deployment on an unseen individual. Reports accuracy, balanced accuracy, macro F1, ROC-AUC.

CNN Test Performance & Training Dynamics

~86%
Multiclass test accuracy (mask sensor)

~95%
Binary test accuracy (mask sensor)

~100%
Binary validation accuracy (converges in <5 epochs)

Failed
Neck oximeter — both tasks failed to generalize

Multiclass Training

Steady, synchronised loss decline through ~15 epochs. Accuracy climbs from ~70% → stable plateau 90–95% on validation set in first 4 epochs. No sign of runaway overfitting.

Binary Training

Simpler problem: validation accuracy hits 100% almost immediately. Loss bottoms near 0.15 within 5 epochs, triggering early stopping automatically.

Neck Oximeter Failure

PPG channels measure secondary cardiovascular responses (SpO_2, pulse rate). Short 10–20 s events produce insufficient O_2 desaturation for a clear CNN signature. CNN overfits patient-specific waveform artifacts instead.

RESULTS — MODEL 2 (ExtraTreesClassifier)

ExtraTreesClassifier LOSO-CV Performance

Metric	Binary Task	Multiclass Task
Accuracy	**88.6%**	**77.9%**
Balanced Accuracy	**87.3%**	**75.8%**
Macro F1	**88.0%**	**76.0%**
ROC-AUC	0.963	0.920 (OvR)
# Estimators	600 ExtraTrees	600 ExtraTrees
Evaluation	LOSO-CV (20 folds)	LOSO-CV (20 folds)

Key insight: Binary task is considerably easier — detecting apnea presence vs. normal. The multiclass task degrades gracefully; dominant confusion is between the two apnea classes (10 s vs. 20 s) which appear similar in their early seconds.

DISCUSSION

CNN vs. ExtraTreesClassifier: Head-to-Head

	Model 1 (CNN)	Model 2 (ExtraTreesClassifier)
Input	Raw time-series (end-to-end)	Hand-crafted statistical + spectral features
Architecture	Stacked 1D Conv + FC heads	ExtraTrees ensemble (600 trees)
Evaluation	Fixed subject split (train/val/test)	LOSO-CV (20 folds — stricter)
Multiclass accuracy	~86%	77.9%
Binary accuracy	~95%	88.6%
Interpretability	Low (black-box)	High (feature importances)
Sensor coverage	Mask only (neck failed)	Both sensors in one pipeline

DISCUSSION

Key Insights & Limitations

Where CNN wins

On inline mask data, the CNN easily learns the mechanical 'flatline' signature of airflow cessation from raw waveforms — near-perfect binary, 86% multiclass.

Where ETC wins

Explicit features (spectral fractions, early-to-late contrast ratios) bypass the high-frequency cardiac noise that confuses the CNN on neck PPG data. Robust across both modalities.

Limitations

- Controlled lab setting — artificially induced apneas; generalizability to clinical OSA unknown
- Only 20 subjects — too small for deep learning; 3-subject test set makes CNN estimate high-variance
- Unequal evaluation: CNN fixed split vs. ETC LOSO-CV prevents a truly fair accuracy comparison
- Algorithmic apnea labels (rolling-std detector), not human-annotated gold-standard intervals
- No SpO_2 computation from raw Red/IR PPG — could improve both models as an explicit feature

Automated Apnea Screening — Outside the Lab

- Both pipelines detect induced apnea with high accuracy — Controlled simulated-apnea detection from low-cost sensors appears feasible.
- CNN excels on direct mechanical signals; ETC is more robust across both modalities
- For resource-constrained deployment, interpretable ExtraTrees is the more practical choice

Future Directions

- Transfer to real sleep datasets (PhysioNet SHHS, MESA) for clinical validation
- Temporal attention (squeeze-excitation / self-attention) to focus CNN on apnea-discriminative segments
- Unified LOSO-CV for both models to enable a statistically fair comparison
- Online/causal streaming inference for real-time wearable deployment